SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 03:39 UTC

SEC P vs NP — daily log 2026-04-28

2 cycles recorded

Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) × Boolean formula depth lower bounds for KRW-style block compositions f ∘ g, the central complexity-theoretic object behind P ⊄ NC^1; equivalently the depth complexity D(f ∘ g) of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded**: 2026-04-28 01:39 UTC
- **Entry ID**: `8047f7f8eea1`

Statement

For all boolean functions f:{0,1}^n→{0,1} and g:{0,1}^m→{0,1} for which HX^1(X_f) and HX^1(X_g) admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace_pairing on the KW-metric spaces obeys the sub-additive bound κ(f ∘ g) ≥ κ(f) + κ(g) − C, where C is a universal constant independent of n,m, and where (X_{f∘g}, d_{f∘g}) is built by the coarse_product operation. In particular, witness cocycles c_f ⊠ c_g ∈ HX^1(X_{f∘g}) and tensor Roe operators T_f ⊗ T_g ∈ C_u^*[X_{f∘g}] realize log|⟨c_f ⊠ c_g, T_f ⊗ T_g⟩| / log(propagation(T_f ⊗ T_g)) ≥ κ(f) + κ(g) − C.

Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 (κ ≤ depth) to recover the KRW conjecture D(f ∘ g) ≥ D(f) + D(g) − O(1). The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: ⟨c_f ⊠ c_g, T_f ⊗ T_g⟩ = ⟨c_f, T_f⟩ · ⟨c_g, T_g⟩ (Connes-style product), while the propagation of T_f ⊗ T_g equals propagation(T_f) · propagation(T_g) under the depth-additive metric of coarse_product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has κ = O(1), so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse_product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
                                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composit
    X_f, d_f = build_X_d(f, k_f)
               ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
         ^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) at build_X_d before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix build_X_d to enumerate vertices correctly (e.g., X = [tuple(b) for b in itertools.product([0,1], repeat=n)]) and rerun the 25-pair × 5-seed protocol.

Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diameter-to-Depth Ratio for Parity-on-a-Tree

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse (Roe) cohomology and geometric group theory applied to circuit complexity × Formula depth lower bounds for the recursive-parity (balanced AND/OR-tree of XORs) function in the NC^1 vs. AC^0[⊕] regime, expressed via the coarse 1-cohomology HX^1(X_f) and the Roe-trace pairing on C_u^*[X_f]
- **Recorded**: 2026-04-28 03:39 UTC
- **Entry ID**: 264c40c0d040

Statement

Let f_n = ⊕-tree_n be the balanced binary tree of XOR gates on n = 2^k inputs, with KW-metric space (X_{f_n}, d_{f_n}). Define the coboundary δ : C^0_coarse(X_{f_n}) -> C^1_coarse(X_{f_n}) and the propagation-R Roe subalgebra C_u^*[X_{f_n}]_R. Then there exists a universal constant α ∈ (0,1) such that, for all R ≤ α·log_2 n, every coarse 1-cocycle c with propagation ≤ R is a coboundary (so its class in HX^1 is trivial), while for R ≥ k = log_2 n there exist nontrivial classes c whose Roe-trace pairing satisfies log|⟨c,T⟩| / log(propagation(T)) ≥ k − O(1). Consequently κ(f_n) ≥ log_2 n − O(1), matching depth(f_n) = Θ(log n) up to constants and certifying axiom A1 sharply for an explicit recursive function.

Rationale

This conjecture directly tests axiom A1 (Coarse-KW link) together with A5 (Roe-index rigidity) on the canonical recursive function whose formula depth is exactly known (depth(⊕-tree_n) = log_2 n).

The KW protocol for the parity tree must descend the tree to find a differing leaf, so the controlled-cover multiplicity must grow by 1 per level; this should manifest as a vanishing-threshold for HX^1 at propagation R ≈ log_2 n, with cocycles at smaller propagation forced to be coboundaries. Verifying both the vanishing below the threshold and the nonvanishing above gives a sharp quantitative version of A1 on a function whose depth we can independently certify, providing a sanity check that κ is neither overshooting nor undershooting the true formula depth in a controlled recursive setting.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1906.11725v2] On the uniform Roe algebra as a Banach algebra and embeddings of ℓ_p uniform Roe algebras

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 73, in run_trial
    tree = xor_tree(n)
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 20, in xor_tree
    left = xor_tree(n // 2)
           ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 22, in xor_tree
    return [left[i] ^ right[i] for i in range(n)]
            ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the xor_tree construction before any data on R*(n), dim HX^1_R, or slope correlations was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the xor_tree recursion (the base case and/or input length handling causing the out-of-range index) and rerun the pre-registered protocol across n ∈ {4,8,16,32}.